



PREPARED FOR DR. SHILPA, KC HOSPITAL, KUPPAM

The Digital Front Desk

An online booking and payment system that lets patients reserve a slot, pay, and receive an instant WhatsApp confirmation — without a staff member touching a phone.

PREPARED

13 August 2026

SCOPE

Booking · Payments · Notifications · Admin

STATUS

Built & running

WHAT THIS IS

A booking desk that never sleeps

Patients pick a date and time on the clinic website, pay the consultation fee online, and get a confirmation on WhatsApp — the same channel they already use every day. No phone tag, no manual entry, no double-booked slots.

BOOKING

Live slot calendar, held for the patient while they pay

PAYMENT

Razorpay checkout, verified two independent ways

CONFIRMATION

WhatsApp + SMS + Email to patient and clinic, automatically

SPEED & RELIABILITY

Built so nothing depends on a person being available

Rather than quote a lab number that means little day-to-day, here is what the system is actually engineered to guarantee — in plain terms.

- **Confirmation fires the instant payment clears**

There is no queue and no staff step in between. The moment Razorpay confirms payment, the system writes the booking as confirmed and sends every message in the same action — typically within seconds of the patient completing checkout.

- **Two patients can never win the same slot**

A slot is held for the paying patient and released automatically if payment isn't completed in time. The final booking is written as one atomic database transition, so even two people checking out at the same second can't both claim 4:00 PM.

- **Payment is confirmed two independent ways**

The booking is confirmed both by the direct response after checkout *and* by a signed webhook Razorpay sends separately. If a patient closes the browser the instant they pay, the webhook still lands the booking — nothing is lost to a dropped connection.

- **One broken channel never blocks another**

WhatsApp, SMS, email and the confirmation call are sent independently. If a carrier rejects one message, it's logged and the rest still go through — a WhatsApp delivery issue can never prevent the booking itself from being confirmed.

- **Every patient gets exactly one confirmation**

Messages fire only on the actual pending → confirmed transition, not on every payment check. A retried or duplicate payment callback — common with online payment gateways — can never send the same patient two confirmations.

WHATSAPP AUTOMATION, VIA TWILIO

From payment to WhatsApp, with nobody in the loop

This is the exact sequence that runs on every paid booking — the same pipeline for the clinic's own notification and the patient's.

1

Patient completes payment

Razorpay checkout returns a signed payment confirmation to the site.

2

Signature verified, slot locked in

The backend checks the payment signature and flips the appointment from "payment pending" to "confirmed" — the step that prevents fake or tampered payment confirmations.

3

Clinic is notified

Dr. Shilpa's team receives the patient's name, phone number, date, time and amount paid — automatically, before the patient even leaves the page.

WhatsApp

SMS

Email

4

Patient is notified

The patient receives their own confirmation with date, time and amount paid, on every channel they might check.

WhatsApp

SMS

Email

TECHNOLOGY

What it's built on, in plain terms

Every layer is a widely-used, actively-maintained platform — nothing exotic that would be hard to hand off, host, or extend later.

WEBSITE

React + Vite

The patient-facing site: fast page loads, works smoothly on mobile, and supports English, Telugu, Tamil and Kannada — including a speak-to-fill voice

input on the appointment form.

SERVER

Python (Flask)

Validates every booking server-side (not just in the browser), manages slot availability, and coordinates payments and notifications.

DATABASE

Supabase (PostgreSQL)

Stores appointments and payments with the same guarantees a bank ledger relies on, so a booking can't be half-written or lost mid-transaction.

PAYMENTS

Razorpay

India-focused payment gateway — cards, UPI, netbanking and wallets — with signature-verified checkout and webhooks.

MESSAGING

Twilio (WhatsApp, SMS & Voice) + Gmail SMTP

The channel that actually reaches patients and staff — WhatsApp and SMS messages and, optionally, an automated confirmation call, alongside email.

ADMIN

Password-protected dashboard

The clinic can see today's and upcoming appointments, confirmed / completed / cancelled / no-show counts, and manage each booking's status.

ALSO INCLUDED

Rounding out the experience

Four languages

English, Telugu, Tamil and Kannada — patients read and book in the language they're comfortable in.

Voice-to-text booking

Patients can speak their appointment request instead of typing, in their chosen language.

Held slots, live countdown

Once a patient starts checkout, their slot is reserved with a visible countdown — no

Mobile-first design

Most patients will book from a phone; the site is built for that first, desktop second.

surprise "already booked" at payment time.

In-dashboard notifications

A notification bell in the admin header flags new activity — bookings, payments, cancellations — right where the clinic team is already looking, no separate app to check.